Evaluating Domain Adaptation in Financial Sentiment Analysis

FinBERT vs. Traditional NLP Baselines

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Motivation and Research Questions

Sentiment analysis is widely used in computational finance to extract market-relevant signals from financial news and corporate disclosures. However, general-domain sentiment models often perform poorly on financial language because word polarity depends strongly on domain context. Terms such as *cost*, *liability*, *debt*, or *depreciation*, for example, carry negative prior polarity in general sentiment resources [2, 6] while serving as neutral descriptive accounting terminology in financial reporting [1, 2].

This project investigates how strongly such **domain mismatch** affects 3-class financial sentiment classification (*positive*, *neutral*, *negative*) and isolates the measurable benefit of financial domain adaptation from general Transformer architecture gains. We address three research questions mapped to our experimental design:

1. **RQ1 (Domain Polarity Mismatch):** To what extent do general-purpose sentiment methods misclassify financial language, particularly by falsely assigning *negative* polarity to *neutral* descriptive statements?
2. **RQ2 (Controlled Architecture vs. Domain Adaptation):** Under a strictly leakage-free evaluation protocol, how much classification improvement stems from domain-adaptive pretraining (FinBERT) versus general-domain BERT-base and traditional machine-learning baselines?
3. **RQ3 (Linguistic Phenomenon Attribution):** Which linguistic phenomena—financial terminology, negation, hedging, numerical comparisons, or context-dependent expressions—drive model divergence?

The main contribution is not only a rigorous, leakage-free comparison of model performance, but also an interpretable phenomenon-level attribution of *when and why* domain adaptation matters in financial NLP.

Data and Experimental Setup

Accessible Datasets. All datasets and lexicons are publicly available and immediately accessible in Python. Our primary in-domain corpus is the **Financial PhraseBank** [1] (Hugging Face `financial_phrasebank`), containing 4,845 financial-news sentences annotated by 16 finance experts as *positive* ($\sim$28%), *neutral* ($\sim$59%), or *negative* ($\sim$13%). To test robustness across annotator consensus tiers, we evaluate on the 100%-agreement subset (`sentences_allagree`, 2,264 sentences), the $\geq$75%-agreement subset (`sentences_75agree`, 3,453 sentences), and the full 4,845-sentence corpus. We pair this with the **Loughran–McDonald Master Dictionary** [2] and an external out-of-distribution (OOD) holdout set, **SEntFiN 1.0** [5] (10,753 entity-sentiment annotations over financial headlines, plus FiQA headlines), filtered to unambiguous single-entity 3-class sentences and deduplicated against PhraseBank via normalized token Jaccard similarity (< 0.8).

Three-Tier Baseline Hierarchy. In Python (`PyTorch`, `transformers`, `scikit-learn`), we compare three levels:

- **Lexicon-based methods (RQ1):** General-domain **VADER** [6] valence rules versus the domain-specific **Loughran–McDonald** financial sentiment lexicon [2], with thresholds tuned strictly on validation splits.
- **Traditional machine learning (RQ1, RQ2): TF-IDF** word and n-gram features with class-weighted **Logistic Regression** and **Linear Support Vector Machines (SVM)** (1–3-grams, `sublinear_tf=True`, `fit_transform` on train folds).
- **Transformer models (RQ2, RQ3):** General-domain **BERT** (`bert-base-uncased` [4]) and finance-adapted **FinBERT** [3], sharing an identical 12-layer architecture to isolate domain pretraining.

Two-Stage Leakage-Free FinBERT Protocol. Off-the-shelf `ProsusAI/finbert` [3] was fine-tuned directly on Financial PhraseBank [1]; evaluating that checkpoint on arbitrary PhraseBank splits leaks training instances into the test set. We eliminate data leakage via a two-stage protocol: (1) **Controlled In-Domain Fine-Tuning:** On Financial PhraseBank, rather than evaluating pre-fine-tuned `ProsusAI/finbert`, we initialize from the **pre-sentiment domain-pretrained FinBERT masked language model** (pretrained only via self-supervised MLM on financial corpora prior to sentiment supervision, e.g., `yiyanghkust/finbert-pretrain` / Araci's MLM checkpoint [3]) and `bert-base-uncased` [4]. Both encoders and the TF-IDF baselines are trained, validated, and tested on the **exact same stratified 70%/15%/15% train/validation/test split** (locked via SHA-256 sentence hashes) and 5-fold stratified cross-validation under identical AdamW settings and validation Macro-F1 early stopping. (2) **External OOD Holdout Check:** To evaluate off-the-shelf `ProsusAI/finbert` alongside

our Stage 1 models without training overlap, we test all models zero-shot on deduplicated **SEntFiN 1.0** [5].

Classification Metrics and Statistical Robustness. Because this is an imbalanced 3-class classification task, we evaluate strictly with classification metrics: **Macro-F1** (primary metric), **Accuracy**, **per-class Precision, Recall, and F1-score** (focusing on false-negative confusion between the *neutral* and *negative* classes), **3×3 Confusion Matrices**, and **Expected Calibration Error (ECE)**. Pairwise differences on identical test splits are verified via paired **McNemar's test** (with continuity correction) and **paired bootstrap resampling** ($B = 10{,}000$ resamples for 95% CIs on ΔMacro-F1).

Error Analysis and Expected Contribution

Phenomenon-Stratified Error Analysis & Attribution (RQ3). Beyond aggregate metrics, we construct a **300-sentence stratified diagnostic subset** from the locked test split. Using deterministic lexical/dependency pre-tagging and double-annotation by two team members (reporting inter-annotator **Cohen's** κ), sentences are grouped into **five linguistic phenomena**: (1) *finance-specific terminology* (e.g., *liability*, *depreciation*, *amortization*), (2) *negation and scope inversion* (e.g., *did not result in a loss*), (3) *hedging and modal uncertainty* (e.g., *may*, *projected to*), (4) *numerical and directional comparisons* (e.g., *margin narrowed from 12% to 9%*), and (5) *context-dependent expressions*. For each category c, we decompose gains into the **architecture effect**, $\Delta_{\text{arch}}(c) = \text{F1}_{\text{BERT-base}}(c) - \text{F1}_{\text{TF-IDF+SVM}}(c)$, and the **domain-adaptation effect**, $\Delta_{\text{domain}}(c) = \text{F1}_{\text{FinBERT-MLM}}(c) - \text{F1}_{\text{BERT-base}}(c)$, isolating domain lexicon mastery from general capacity.

Reusable Value Produced. We will release three open-source deliverables on GitHub: (i) **leakage-free benchmark split manifests** (SHA-256 sentence hashes and PhraseBank–SEntFiN deduplication scripts), (ii) the **300-sentence phenomenon-annotated financial sentiment diagnostic dataset** with annotation guidelines, and (iii) a **reproducible Python evaluation pipeline** running all baselines, statistical tests, and error analyses.

Work Plan

The project is scoped for a **55-hour total team budget** (∼18.3h per member across Qiyao Zheng, Benjamin Donick, and Necati Furkan Colak) structured into five work packages:

- **WP1 — Data & Leakage-Free Splits (8h; Lead: Colak):** Ingest PhraseBank (`allagree`, `75agree`, full 4,845) and SEntFiN 1.0; lock SHA-256 70/15/15 and 5-fold stratified splits; run Jaccard (< 0.8) filtering.
- **WP2 — Lexicon & TF-IDF Baselines (10h; Lead: Donick):** Implement VADER [6] and Loughran–McDonald [2] classifiers; train leakage-free TF-IDF (1–3-gram) + Logistic Regression and Linear SVM.
- **WP3 — BERT vs. FinBERT Fine-Tuning (14h; Lead: Zheng):** Fine-tune `bert-base-uncased` [4] and MLM FinBERT (`yiyanghkust/finbert-pretrain` [3]); test `ProsusAI/finbert` on SEntFiN 1.0 [5].
- **WP4 — Phenomenon Annotation & Statistical Analysis (13h; All):** Double-annotate the 300-sentence diagnostic set across the 5 phenomena (Cohen's κ); compute 3×3 confusion matrices, ECE, McNemar's test, 10,000-sample bootstrap CIs, and Δ_{arch} vs. Δ_{domain} attribution.
- **WP5 — Reproducible Repository & Final Report (10h; All):** Package CLI scripts, split manifests, and diagnostic dataset on GitHub; write and proofread the final report.

References

[1] P. Malo, A. Sinha, P. Korhonen, J. Wallenius, and P. Takala, "Good Debt or Bad Debt: Detecting Semantic Orientations in Economic Texts," *Journal of the Association for Information Science and Technology*, vol. 65, no. 4, pp. 782–796, 2014.

[2] T. Loughran and B. McDonald, "When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks," *The Journal of Finance*, vol. 66, no. 1, pp. 35–65, 2011.

[3] D. Araci, "FinBERT: Financial Sentiment Analysis with Pre-trained Language Models," *arXiv preprint arXiv:1908.10063*, 2019.

[4] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proceedings of NAACL-HLT*, pp. 4171–4186, 2019.

[5] A. Sinha, S. Kedas, R. Kumar, and P. Malo, "SEntFiN 1.0: Entity-Aware Sentiment Analysis for Financial News," *J. Assoc. Inf. Sci. Technol.*, vol. 73, no. 9, pp. 1314–1335, 2022.

[6] C. J. Hutto and E. Gilbert, "VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text," in *Proc. Int. AAAI Conf. Web and Social Media (ICWSM)*, vol. 8, no. 1, pp. 216–225, 2014.